POWER PULSE

Circuit Breakers Important Formulae

Notes by Balaji Sir

1 Ratings

1. $\text{Breaking MVA} = \sqrt{3}\, V_L(\text{kV})\, I_b(\text{kA})$ (symmetrical rms at rated voltage)
2. $\text{Making current} \approx 2.55 \times I_{b,sym}\ (\text{peak}), \qquad I_{sc} = \dfrac{MV\,A_{base}}{\sqrt{3}\,V\,X_{pu}}$
3. Rated sequence: $O - 0.3\ \text{s} - CO - 3\ \text{min} - CO$

2 Transient recovery voltage (TRV)

4. $\omega_n = \dfrac{1}{\sqrt{LC}}, \quad f_n = \dfrac{1}{2\pi\sqrt{LC}}$
5. $v(t) = E_m\,(1 - \cos\omega_n t), \qquad v_{peak} = 2E_m$

E_m = peak phase voltage (undamped)

6. $RRRV_{max} = E_m\,\omega_n \ \ (\text{at } \omega_n t = \tfrac{\pi}{2})$
7. $R_{critical} = \dfrac{1}{2}\sqrt{\dfrac{L}{C}}$ (resistance switching: no oscillation for $R \le R_c$)

parallel resistor across contacts

3 Overvoltages

8. $\tfrac{1}{2}LI^2 = \tfrac{1}{2}CV^2 \ \Rightarrow\ V = I\sqrt{L/C}$

current chopping (inductive switching)

9. Capacitive switching: restrike can give up to $3\times$ the peak voltage

4 Types

10. ACB, oil (bulk/minimum), air-blast, vacuum, SF_6 (best for EHV)
11. Arc extinction at current zero: high-pressure cooling, high resistance, high-dielectric gas